Project Documentation

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MCD | MAESTRÍA EN CIENCIA DE LOS DATOS

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Challenge_Basic

Glassdoor Companies Feedback Analysis (NLP & MLOps)

1. Project Description

This project aims to analyze employee feedback (Pros and Cons) from Glassdoor using Natural Language Processing (NLP) techniques. The pipeline includes automated web scraping data handling, sentiment analysis for both English and Spanish, and an MLOps integration for experiment tracking.

2. Dataset Extraction

The analysis is based on the employee_reviews.csv dataset

Source: Real-world company reviews from Glassdoor.

Content: 67,529 entries featuring text-based feedback and numerical ratings.

Preprocessing: The data is cleaned by removing irrelevant variables, handling missing values, and splitting the corpus by language (English and Spanish).

3. Model Construction

The sentiment analysis follows a comparative approach:

English Corpus: Evaluated using VADER (Valence Aware Dictionary and sEntiment Reasoner) due to its efficiency with social media-style text.

Spanish Corpus: Evaluated using pysentimiento, a specialized library for Spanish NLP tasks.

Feature Engineering: Includes Lemmatization, Stop-word removal, and N-gram distribution analysis.

4. MLOps Integration

Experiment tracking is managed via MLflow.

Metrics: Logging sentiment distribution and positive review rates.

Artifacts: Storage of model signatures, sentiment plots, and N-gram frequency charts.

Tracking: All runs are logged to a local server for future reference.

5. Execution Guide

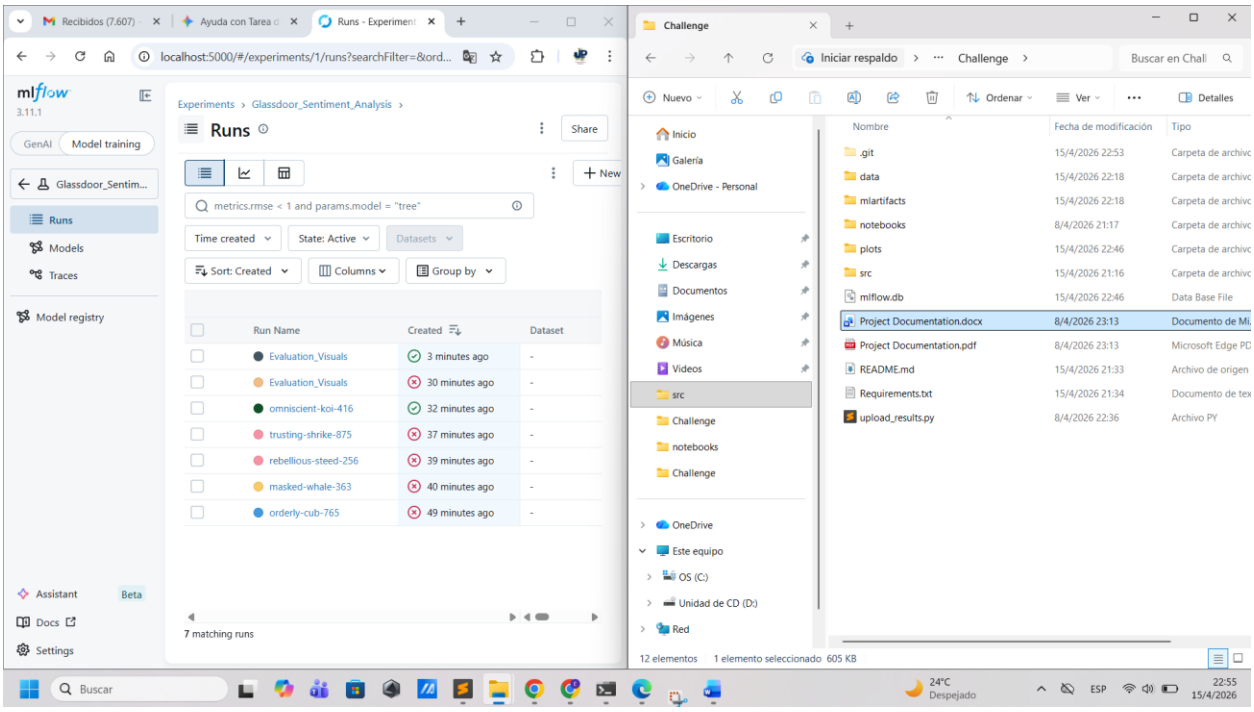
To run this project, follow these steps:

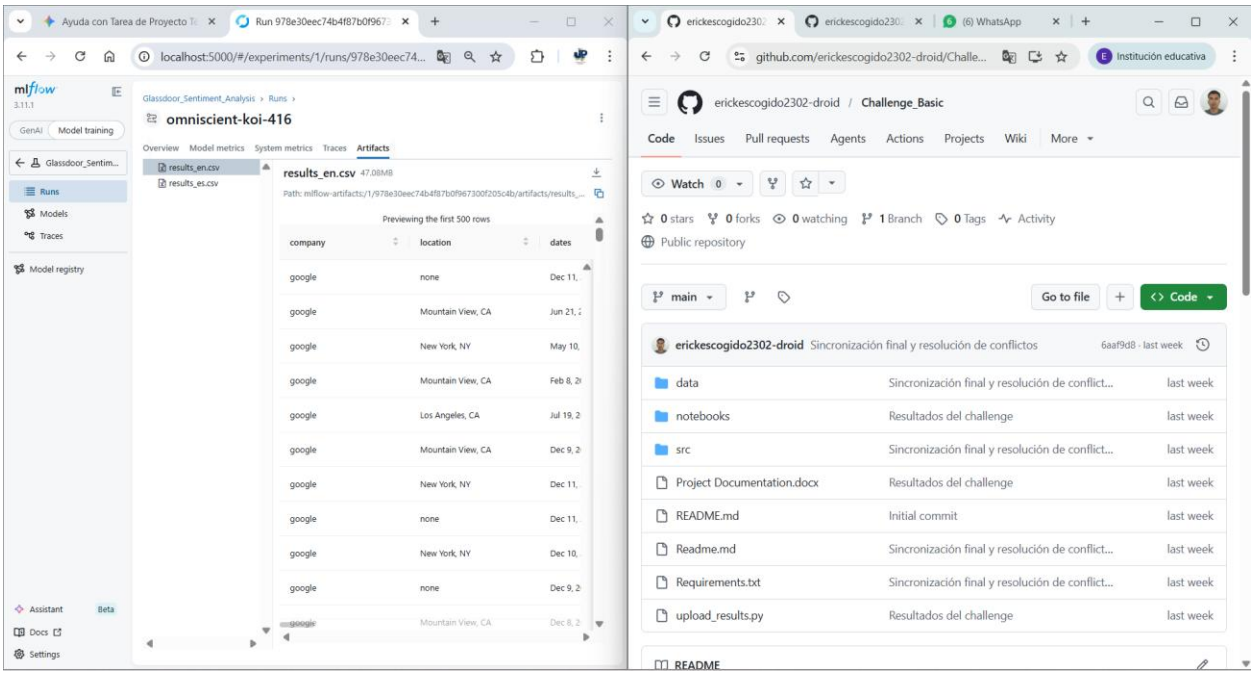
Prerequisites

Python 3.x

MLflow server running locally (mlflow ui)

Evidence.

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mlflow
3.11.1
Glassdoor_Sentiment_Analysis › Runs ›
omniscient-koi-416
Overview
Model metrics
System metrics
Traces
Artifacts
results_en.csv
results_es.csv
results_en.csv 47.08MB
Previewing the first 500 rows
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location
dates
google
none
Mountain View, CA
New York, NY
Los Angeles, CA
Runs
Models
Traces
Model registry
Assistant
Beta
Docs
Settings
github.com/erickescogido2302-droid/Challe...
erickescogido2302-droid / Challenge_Basic
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
More
Watch 0
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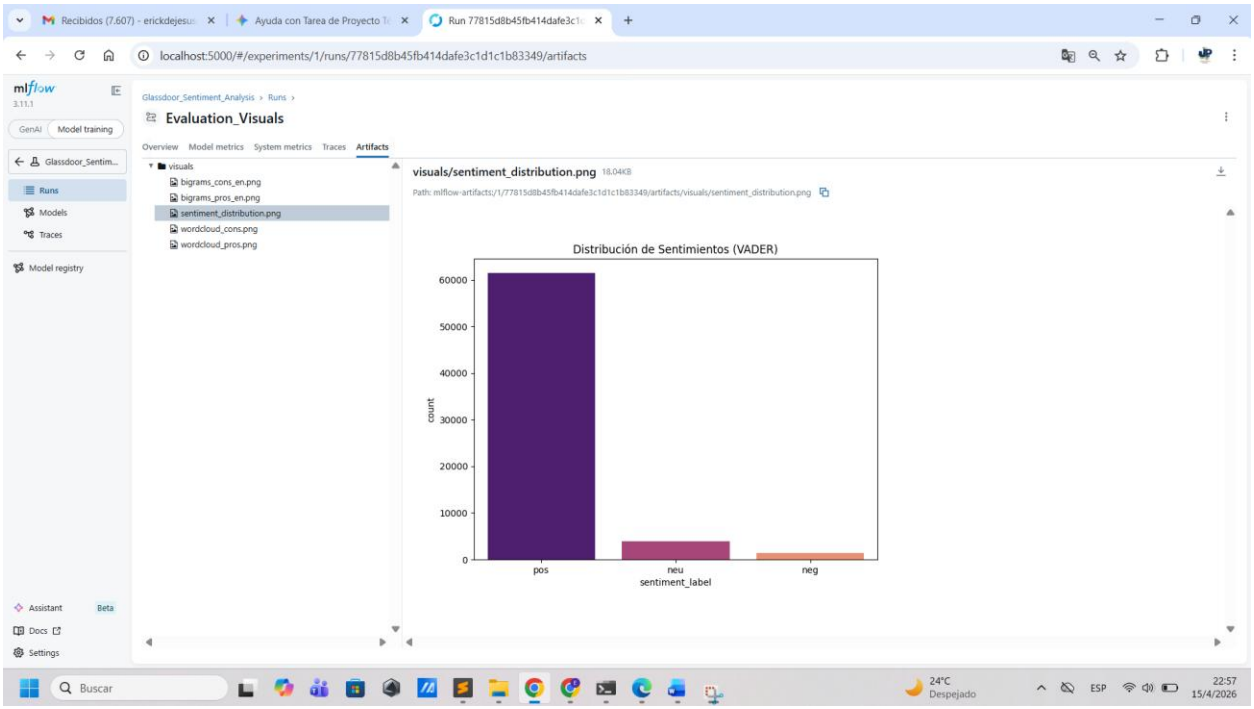
0 watching
1 Branch
0 Tags
Activity
Public repository
main
Go to file
Code
erickescogido2302-droid Sincronización final y resolución de conflictos
6aaf9d8 · last week
data
notebooks
src
Project Documentation.docx
README.md
Readme.md
Requirements.txt
upload_results.py
Resultados del challenge
Initial commit
last week
README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mlflow
3.11.1
Glassdoor_Sentiment_Analysis › Runs ›
Evaluation_Visuals
Overview
Model metrics
System metrics
Traces
Artifacts
visuals
bigrams_cons_en.png
bigrams_pros_en.png
sentiment_distribution.png
wordcloud_cons.png
wordcloud_pros.png
visuals/sentiment_distribution.png 18.04KB
Distribución de Sentimientos (VADER)
count
60000
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30000
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Runs
Models
Traces
Model registry
Assistant
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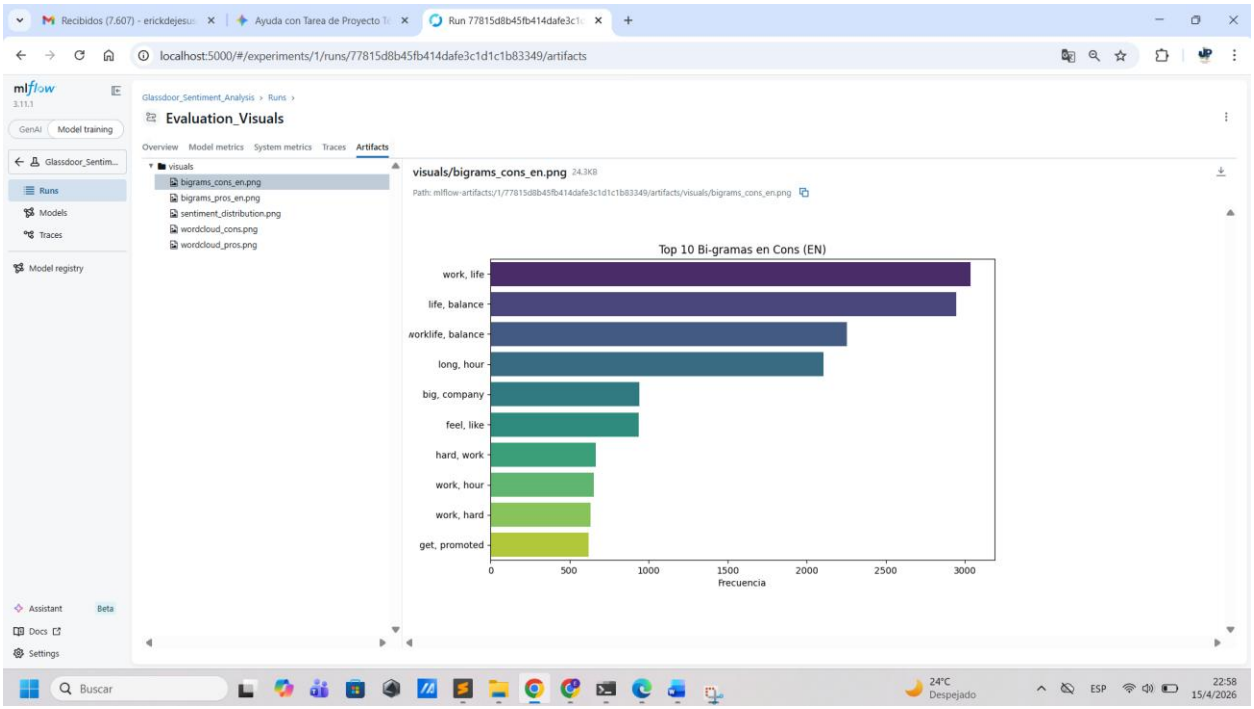
localhost:5000/#/experiments/1/runs/77815d8b45fb414dafe3c1d1c1b83349/artifacts
Glassdoor_Sentiment_Analysis › Runs ›
Evaluation_Visuals
Overview
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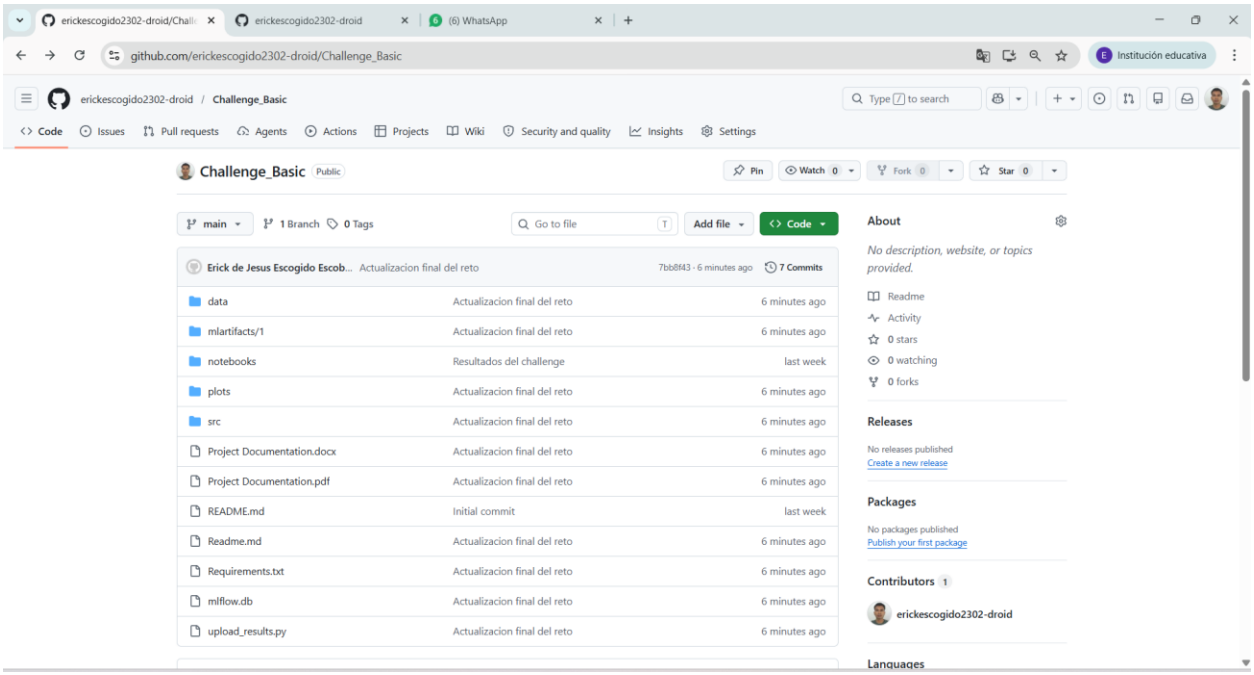
visuals
bigrams_cons_en.png
bigrams_pros_en.png
sentiment_distribution.png
wordcloud_cons.png
wordcloud_pros.png
visuals/bigrams_cons_en.png 24.3KB
Top 10 Bi-gramas en Cons (EN)
work, life
life, balance
worklife, balance
long, hour
big, company
feel, like
hard, work
work, hour
work, hard
get, promoted
0
500
1000
1500
2000
2500
3000
Frecuencia


github.com/erickescogido2302-droid/Challenge_Basic
erickescogido2302-droid / Challenge_Basic
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Challenge_Basic Public
main
1 Branch
0 Tags
Go to file
Add file
Code
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7bb8f43 · 6 minutes ago
7 Commits
data
mlartifacts/1
notebooks
plots
src
Project Documentation.docx
Project Documentation.pdf
README.md
Readme.md
Requirements.txt
mlflow.db
upload_results.py
About
No description, website, or topics provided.
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
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Languages